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OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

LECTURE 28: THE QUEUING SYSTEM $M/M/1/GD/c/\infty$

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Recall, from Lecture 26, that

$$\begin{aligned} P_{i,j}(t + \Delta t) = & \Delta t[\lambda_{j-1}P_{i,j-1}(t) + \mu_{j+1}P_{i,j+1}(t) - \mu_j P_{i,j}(t) - \lambda_j P_{i,j}(t)] \\ & + P_{i,j}(t) + o(\Delta t) \end{aligned}$$

Letting

$$\begin{aligned} \lambda_j &= \lambda, \quad j = 0, 1, \dots, c-1 \\ \mu_j &= \mu, \quad j = 1, 2, \dots, c \\ \lambda_c &= \mu_0 = 0, \end{aligned}$$

this leads to a system of differential equations. Letting $P_{i,j}(t) = \pi_j$ in the steady state as $t \rightarrow \infty$ and noting that $\pi'_j = 0$, this system of differential equations reduces to a system of algebraic equations which can be solved for the two cases $\lambda \neq \mu$ and $\lambda = \mu$ separately.

The System $M/M/1/GD/c/\infty$ when $\lambda \neq \mu$ (p. 1083)

For $\rho = \frac{\lambda}{\mu}$ and $\lambda \neq \mu$ (i.e. $\rho \neq 1$):

$$\pi_j = \begin{cases} \frac{1-\rho}{1-\rho^{c+1}}, & j = 0 \\ \rho^j \pi_0, & j = 1, 2, \dots, c \\ 0, & j = c+1, c+2, \dots \end{cases}$$

$$L = \frac{\rho[1 - (c+1)\rho^c + c\rho^{c+1}]}{(1 - \rho^{c+1})(1 - \rho)}$$

$$L_s = 1 - \pi_0$$

$$L_q = L - L_s$$

$$W = \frac{L}{\lambda(1 - \pi_c)}$$

$$W_q = \frac{L_q}{\lambda(1 - \pi_c)}$$

$$W_s = \frac{L_s}{\lambda(1 - \pi_c)}$$

The System $M/M/1/GD/c/\infty$ when $\lambda = \mu$ (p. 1083)

For $\rho = \frac{\lambda}{\mu}$ and $\lambda = \mu$ (i.e. $\rho = 1$):

$$\pi_j = \begin{cases} \frac{1}{c+1}, & j = 0, 1, \dots, c \\ 0, & j = c+1, c+2, \dots \end{cases}$$

$$L = \frac{c}{2}$$

$$L_s = \frac{c}{c+1}$$

$$L_q = L - L_s$$

$$W = \frac{L(c+1)}{\lambda c}$$

$$W_q = \frac{L_q(c+1)}{\lambda c}$$

$$W_s = \frac{L_s(c+1)}{\lambda c}$$

Example 8 (p. 1084)

A one-man barber shop has a total of 10 seats.

Interarrival times are exponentially distributed, and an average of 20 prospective customers arrive each hour at the shop.

Those customers who find the shop full do not enter.

The barber takes an average of 12 minutes to cut each customer's hair, and haircut times are exponentially distributed.

- 1 On average, how many haircuts per hour will the barber complete?
- 2 On average, how much time will be spent in the shop by a customer who enters?